The humans who **performed BIG-bench tasks**, and whose scores are reported in this paper, are Vinay Ramasesh, Daniel Freeman, Vedant Misra, and an expert evaluation team consisting of Andrew La, Angela Jiang, Stephen Prasad, Joseph Boudeman, Joseph Guerr, Joyce Chua, Melvin McElrath, Ryan Stovall, Melody Arnaud, Katerina Ignatyeva, Tyler Shultz, Michael Starritt, Sean Casey, Cindy Ramirez, Hayden Bogar, Anh Vuong, Cameron Dour, Michael Gu, and Theodore Rothschild, and managed by Courtney Ashcraft.

The **initial project idea** was developed in conversation between Guy Gur-Ari, Ethan Dyer, and Jascha Sohl-Dickstein. The initial design doc for the project was written by Jascha Sohl-Dickstein, and further developed by Guy Gur-Ari, Ethan Dyer, Daniel Freeman, Aitor Lewkowycz, and others.

A Google-internal **"call for tasks" project** was created by Noah Fiedel and Gaurav Mishra and then merged into BIG-bench. This "call for tasks" project is the source of the JSON task API in BIG-bench.

BIG-bench was **managed and organized** by Guy Gur-Ari, Jascha Sohl-Dickstein, Noah Fiedel, and Ethan Dyer.

The **BIG-bench paper** was written by Jascha Sohl-Dickstein, Guy Gur-Ari, Ethan Dyer, Daniel Freeman, Noah Fiedel, Vinay Ramasesh, Anders Andreassen, Jaehoon Lee, Noah Constant, Rosanne Liu, Aitor Lewkowycz, Vedant Misra, and Ambrose Slone.

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Meta-review of task submissions was performed by Vinay Ramasesh, Jascha Sohl-Dickstein, Guy Gur-Ari, Ethan Dyer, Anders Andreassen, Aitor Lewkowycz, Daniel Freeman, Jaehoon Lee, Vedant Misra, Rosanne Liu, Noah Fiedel, Gaurav Mishra, and Ambrose Slone. Vinay Ramsesh, Kristen Chiafullo, Guy Gur-Ari, and Jascha Sohl-Dickstein acted at various times as meta-meta reviewers, and coordinated the review process.

Review of BIG-bench tasks was performed by Abhishek Rao, Adam Brown, Aditya Gupta, Adrià Garriga-Alonso, Aitor Lewkowycz, Akshat Agarwal, Alethea Power, Alicia Parrish, Allen Nie, Ambrose Slone, Anders Andreassen, Andrea Madotto, Andrew Dai, Anna Golubeva, Avia Efrat, Behnam Hedayatnia, Behnam Neyshabur, Behrooz Ghorbani, Boris Hanin, Chandan Singh, Charles Rathkopf, Colin Raffel, Daniel Freeman, Daniel Khashabi, Dar Gilboa, David Dohan, Denis Emelin, Derek Tam, Ernie Chang, Ethan Dyer, Gaurav Mishra, Genta Indra Winata, Gerard de Melo, Gregor Betz, Guy Gur-Ari, Harsh Mehta, Hongming Zhang, Igor Babuschkin, Ilan Buzzetti, Isaac Noble, Jack Geissinger, Jaehoon Lee, James B. Simon, James Koppel, James Sully, Jascha Sohl-Dickstein, Jekaterina Novikova, Jesse Michael Han, Joshua Rule, Karthik Gopalakrishnan, Kaustubh D. Dhole, Kyle Aitken, Ludwig Schmidt, Luke Metz, Marcelo Menegali, Marco Maru, Michele Bevilacqua, Michihiro Yasunaga, Mor Geva, Mozhdeh Gheini, Nayeon Lee, Neha Wadia, Nicholas Roberts, Nick Doiron, Nitish Shirish Keskar, Noah Fiedel, Paul Liang, Qing Lyu, Richard Barnes, Rosanne Liu, Samuel Schoenholz, Shixiang Shane Gu, Shubham Toshniwal, Shyam Upadhyay, Sneha Makini, Stanislav Fort, Stella Biderman, Tal Linzen, Tal Schuster, Tao Yu, Thomas Phan, Timothy Nguyen, Vedant Misra, Vikas Raunak, Vinay Prabhu, Vinay Ramasesh, William Saunders, Xudong Shen, Zijian Wang, and Zijie Jay Wang.

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7.2 Task authors, and task-specific citations

`abstract_narrative_understanding`: Sayan Ghosh and Shashank Srivastava, UNC Chapel Hill (sayghosh, ssrivastava)@cs.unc.edu. Task cites: (Ghosh & Srivastava, 2021; Holyoak, 2012; Nippold et al., 2001; Tan et al., 2015; Wang et al., 2020b; Mostafazadeh et al., 2016)

`abstraction_and_reasoning_corpus`: Evgenii Zheltonozhskii, Technion – Israel Institute of Technology (evgeniizh@campus.technion.ac.il), and Francois Chollet, Google (fchollet@google.com). Task cites: (Chollet, 2019; Brown et al., 2020; Chollet, 2020)

anachronisms: Peter Chang, Mike Cain, Spencer Torene, Eleanor Hagerman, Zachary Seid, Berk Ekmekci, and Blake Howald, Thomson Reuters Special Services (peter.chang, michael.cain, s.torene, eleanor.hagerman, zach.seid, berk.ekmekci, blake.howald)@thomsonreuters.com. Task cites: (Otterbacher et al., 2002; Popescu & Strapparava, 2015; Llorens et al., 2015; Meng et al., 2017; Geva et al., 2021)

analogical_similarity: Denis Kleyko, UC Berkeley and Research Institutes of Sweden (denkle@berkeley.edu), and Cameron Diao, Rice University (cwd2@rice.edu). Task cites: (Plate, 2003; 1994; Thagard et al., 1990; Gentner et al., 1993)

analytic_entailment: Rachel Etta Rudolph, Auburn University (rer0037@auburn.edu); Alexander W. Kocurek, Cornell University (awk78@cornell.edu); and Ethan Jerzak, National University of Singapore (phiejj@nus.edu.sg). Task cites: (Hume, 1739–1740; Kant, 1781/1787; Wittgenstein, 1953; Quine, 1951; Grice & Strawson, 1956; Bolukbasi et al., 2016; Kocurek et al., 2020; Rudolph & Kocurek, 2020; Kocurek & Jerzak, 2021)

arithmetic: Guy Gur-Ari, Google (guyga@google.com). Task cites: (Brown et al., 2020; Saxton et al., 2019)

ascii_word_recognition: Nitish Shirish Keskar, Salesforce (nkeskar@salesforce.com). Task cites: (Child et al., 2019; Chen et al., 2020)

authorship_verification: Niklas Deckers and Martin Potthast, Leipzig University and ScaDS.AI (niklas.deckers, martin.potthast)@uni-leipzig.de; Sebastian Bischoff, Leipzig University, ScaDS.AI and University of Tübingen (sebastian@salzreute.de); Benno Stein, Bauhaus-Universität Weimar (benno.stein@uni-weimar.de); and Matthias Hagen, Friedrich-Schiller-Universität Jena (matthias.hagen@uni-jena.de). Task cites: (Bischoff et al., 2020; Koppel & Schler, 2004)

auto_categorization: Fanyue Xia, Carnegie Mellon University (fanyuex@andrew.cmu.edu)

auto_debugging: Mo Tiwari, Medina Orduna Baitemirova, and Chris Waites, Stanford University (motiwari, medinab, waites)@stanford.edu. Task cites: (Zaremba & Sutskever, 2014)

bbq_lite: Alicia Parrish, Angelica Chen, Phu Mon Htut, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jana Thompson, and Samuel R. Bowman, New York University (alicia.v.parrish, ac5968, pmh330, nikitanangia, vp1271, zp489, jt4407, bowman)@nyu.edu. Task cites: (Crawford, 2017; Khashabi et al., 2020b; Li et al., 2020a)

bbq_lite_json: Alicia Parrish, Angelica Chen, Phu Mon Htut, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jana Thompson, and Samuel R. Bowman, New York University (alicia.v.parrish, ac5968, pmh330, nikitanangia, vp1271, zp489, jt4407, bowman)@nyu.edu

bias_from_probabilities: Yasaman Bahri and Jascha Sohl-Dickstein, Google (yasamanb, jaschasd)@google.com. Task cites: (Bender et al., 2021; Abid et al., 2021)

boolean_expressions: Roman Novak, Google (romann@google.com). Task cites: (Habernal et al., 2018; Yu et al., 2020; Dua et al., 2019; Liu et al., 2020a; Sinha et al., 2019; Wang et al., 2019a; Steinbach & Kohut, 2002; Saxton et al., 2019; Payani & Fekri, 2019; Trask et al., 2018; Selsam et al., 2018; Allamanis et al., 2016; Evans et al., 2018; Shi et al., 2020)

bridging_anaphora_resolution_barqa: Yufang Hou, IBM Research Europe, Ireland (bnuxiaofang@gmail.com); Katja Markert, Heidelberg University (markert@cl.uni-heidelberg.de); and Michael Strube, Heidelberg Institute for Theoretical Studies (michael.strube@h-its.org). Task cites: (Hou, 2020; Hou et al., 2013; Markert et al., 2012; Rajpurkar et al., 2016)

causal_judgment: Allen Nie, Tatsu Hashimoto, and Tobias Gerstenberg, Stanford University (anie, thashim, gerstenberg)@stanford.edu. Task cites: (Gordon, 2010; Bosselut et al., 2019; Halpern, 2016; Knobe, 2003)

cause_and_effect: Guy Gur-Ari, Google (guyga@google.com), and Neta Gur-Ari Krakover (krakover@gmail.com). Task cites: (Gordon, 2010)

checkmate_in_one: Nitish Shirish Keskar, Salesforce (nkeskar@salesforce.com). Task cites: (Alexander, 2020; Ammanabrolu et al., 2019; Dambekodi et al., 2020; Ammanabrolu et al., 2020)

chess_state_tracking: Karen Livescu, Kevin Gimpel, and Shubham Toshniwal, TTI Chicago (klivescu, kgimpel, shtoshni)@ttic.edu; and Sam Wiseman, Duke University (swiseman@cs.duke.edu). Task cites: (Weston et al., 2015; Côté et al., 2018; Toshniwal et al., 2021; Alexander, 2020; Chen, 2020; Noever et al., 2020; Swingle et al., 2021)

chinese_remainder_theorem: Jiaming Song, Chenlin Meng, and Stefano Ermon, Stanford University (jiaming.tsong, chenlin, ermon)@cs.stanford.edu

cifar10_classification: Peiyuan Liao, Carnegie Mellon University, Praxis Pioneering (peiyuanl@cs.cmu.edu)

code_line_description: Rabin Banjade and Priti Oli, University of Memphis (rbnjade1, poli)@memphis.edu. Task cites: (Alon et al., 2018)

codenames: Isaac Noble and Lucy Noble, Google (isaacn, lucynoble)@google.com, Emma Lam (emmas.email1720@gmail.com), and Lucas Lam (lucas06lam@gmail.com). Task cites: (Kim et al., 2019)

color: Zijie J. Wang, Georgia Tech (jayw@gatech.edu). Task cites: (Gibson et al., 2017; Zucconi, 6 Jan. 2016)

com2sense: Shikhar Singh, Nuan Wen, and Yu Hou, University of Southern California (ssingh43, nuanwen, houyu)@usc.edu; Pegah Alipoormolabashi, Sharif University of Technology (palipoor976@gmail.com); and Te-Lin Wu and Nanyun Peng, University of California, Los Angeles (UCLA) (telinwu,violetpeng)@cs.ucla.edu. Task cites: (Singh et al., 2021)

common_morpheme: Shyamolima (Shammie) Debnath, University of Washington (shyamolima.d@gmail.com). Task cites: (Devlin et al., 2018; Wu et al., 2016; Won et al., 2021; Xu et al., 2018; Edmiston & Stratos, 2018; El-Kishky et al., 2019)

conceptual_combinations: Raphaël Millière, Columbia University (rm3799@columbia.edu); Catherine Stinson, Queen's University at Kingston (c.stinson@queensu.ca); Dimitri Coelho Mollo, Umeå University (dimitri.mollo@umu.se); and Charles Rathkopf, Juelich Research Center (c.rathkopf@fz-juelich.de). Task cites: (Fodor, 1975; Fodor & Pylyshyn, 1988; Smolensky, 1988; Lake et al., 2017; Lake & Murphy, 2020; Marcus, 2020; Henrich et al., 2010; Murphy, 1988)

conlang_translation: James Koppel, MIT (jkoppel@mit.edu); Rowan Jacobs, Synthego (rowan.jacobs@synthego.com); and Mimee Xu, NYU (mimee@nyu.edu). Task cites: (Canfield, 2010; Şahin et al., 2020; Sennrich & Zhang, 2019)

context_definition_alignment: Lütfi Kerem Şenel, Hinrich Schütze, and Timo Schick, LMU Munich (lksenel@gmail.com, inquiries@cislmu.org, and schickt93@gmail.com). Task cites: (Senel & Schütze, 2021; Reimers & Gurevych, 2019)

convinceme: Cedrick Argueta and Jaime Fernández Fisac, Princeton University (cedrick, jfisac)@princeton.edu; and Vinay Ramasesh, Google (ramasesh@google.com). Task cites: (Lin et al., 2021b; Levy et al., 2021; Clark et al., 2018; Maynez et al., 2020; Wang et al., 2020a; Kenton et al., 2021; Xu et al., 2020a; Tamkin et al., 2021)

coqa_conversational_question_answering: Siva Reddy, Mila and McGill University (siva.reddy@mila.quebec); Danqi Chen, Princeton University (danqic@cs.princeton.edu); and Christopher D. Manning, Stanford University (manning@stanford.edu). Task cites: (Reddy et al., 2019; Radford et al., 2019; Brown et al., 2020)

crash_blossom: Amanda Dsouza, Georgia Tech (dsouza.amanda@gmail.com); Matthew L. Leavitt, MosaicML (matthew.l.leavitt@gmail.com); Parth Doshi, UC San Diego (pdoshi@ucsd.edu); and Animesh Gupta, Thapar University (animeshgupta.thapar@gmail.com)

crass_ai: Jörg Frohberg, Apergo UG (j.frohberg@apergo.ai). Task cites: (Schölkopf et al., 2021; Teney et al., 2020; Liang et al., 2020b; Pearl, 2000; Shahid & Zheleva, 2021; Xia et al., 2021; Priol et al., 2020)